Algorithm 1 Meta-evolution

Inputs: baseline; training tasks T; validation tasks T_{val}; G generations; O offspring per generation; P population size; N_{init} initial seeds; N_{val} validation seeds; reevaluation `strategy`: either `TTTS` with B seeds per generation, or `uniform` with N_{reeval} seeds per new population member.

EVALUATE(`baseline`, N_{init})
`archive` $\leftarrow$ {`baseline`}
`population` $\leftarrow$ [`baseline`]
for $g = 1, \ldots, G$ **do**
  for $o = 1, \ldots, O$ **do** ▷ *Sample offspring*
    `parents` $\sim$ SAMPLEPARENTS(`population`)
    `prompt` $\sim$ PROMPT($T, g,$ `parents`)
    $x_o \sim$ LLM(`prompt`)
    EVALUATE(x_o, N_{init})
    `archive` $\leftarrow$ `archive` $\cup \{x_o\}$
  `population` $\leftarrow \{\arg\max_{x \in \texttt{archive}} \hat{\mu}_x(t) \mid t \in T\}$ ▷ *MAP-Elites, one niche per task*
  Fill remaining slots up to P by highest $\hat{\mu}_x$
  if `strategy` = `TTTS` **then** ▷ *Reevaluate*
    $\hat{\sigma} \leftarrow$ pooled standard deviation of per-seed scores over `archive`
    $\psi_x \leftarrow$ top-two Thompson sampling probability of x under posterior $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$
    $k \sim \text{Multinomial}(B, \psi)$
    EVALUATE(x, k_x) for each $x \in$ `archive`
  else if `strategy` = `uniform` **then**
    EVALUATE(x, N_{reeval}) for each new member x of `population`
  $x_g^* \leftarrow \arg\max_{x \in \texttt{population}} \hat{\mu}_x$ ▷ *Validate generation's best*
  `val`$[x_g^*] \leftarrow$ mean score of x_g^* on T_{val} over N_{val} seeds
return $\arg\max_x$ `val`$[x]$

function EVALUATE(x, n)
  Run x on each task in T with n new seeds, adding the per-seed scores to S_x
  $\hat{\mu}_x, n_x \leftarrow$ mean and size of S_x
  $\hat{\mu}_x(t) \leftarrow$ mean of S_x on task t